XAI-NLG: A Faithfulness-Constrained Natural Language Generation Pipeline for Explainable Customer Churn Prediction

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Abstract

Customer churn prediction has emerged as a critical application of machine learning in the telecommunications industry. While high-performing models such as XGBoost and Artificial Neural Networks (ANNs) achieve strong predictive accuracy, they operate as black-box systems whose internal decision logic is inaccessible to business stakeholders. Existing post-hoc explainability methods — most notably SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations) — address this opacity by producing feature attribution scores, yet two critical gaps persist: *(1) there is no systematic evaluation of the degree to which SHAP and LIME agree with each other across different model architectures, (2) raw attribution outputs remain technically opaque to non-ML practitioners and do not directly support business decision-making.*

This paper presents **XAI-NLG**, an end-to-end pipeline that addresses both gaps. We train four machine learning models — Logistic Regression, Random Forest, XGBoost, and ANN — on the standard Telco Customer Churn dataset and evaluate their explanations along four axes: **faithfulness**, **stability**, **cross-method consistency**, and **business readability**. We introduce the **Explanation Stability Index (ESI)** and a formal SHAP–LIME rank agreement metric, constituting the first comparative explainability evaluation on this benchmark. We further propose a **faithfulness-constrained LLM prompt schema** that grounds OpenAI-generated business narratives in verified SHAP attribution vectors, accompanied by a rule-based hallucination detection pipeline. The Artificial Neural Network achieves the highest AUC (0.835), but XGBoost represents the optimal accuracy–interpretability Pareto point, combining competitive AUC with significantly higher explanation stability. Constrained prompting reduces attribution hallucination rates relative to unconstrained generation, and human evaluators rate LLM narratives significantly higher on actionability than raw XAI visualizations. The system is deployed as an interactive Streamlit dashboard providing per-customer churn narratives and retention recommendations.

Keywords: explainable AI, customer churn prediction, SHAP, LIME, large language models, natural language generation, faithfulness, explanation stability, human-centered AI, trustworthy AI

1. Introduction

Customer churn — the phenomenon of customers discontinuing a service — represents one of the most economically significant challenges in subscription-based industries. Telecommunication companies, banking institutions, streaming services, and Software-as-a-Service platforms collectively lose hundreds of billions of dollars annually to preventable attrition. Predictive models that identify at-risk customers before they churn enable proactive retention strategies, reducing acquisition costs and improving customer lifetime value.

Machine learning has become the dominant paradigm for churn prediction. Gradient-boosted models such as XGBoost and deep neural networks consistently achieve high accuracy on structured customer data, and their adoption in production systems has grown substantially. However, these high-performing models operate as black boxes: they produce a churn probability but provide no human-interpretable justification for that prediction. In business environments where managers must decide whether to offer a discount, escalate a case to a retention specialist, or change a customer's service plan, a probability score without an explanation is insufficient for confident action.

The field of Explainable Artificial Intelligence (XAI) has produced post-hoc attribution methods — most prominently SHAP (Lundberg & Lee, 2017) and LIME (Ribeiro et al., 2016) — that decompose model predictions into per-feature contribution scores. These techniques have been widely applied to churn prediction, and their outputs have demonstrably improved model trust among ML practitioners. However, two substantive research gaps remain.

First, there is no systematic cross-architecture comparison of SHAP and LIME agreement. The two methods operate under different theoretical assumptions (global Shapley coalition averaging vs. local linear approximation), and their outputs may diverge substantially for complex models like ANNs. If a stakeholder is told that 'contract type' is the top churn driver by SHAP but LIME attributes that prediction primarily to 'monthly charges', the choice of explainability method has produced materially different and potentially contradictory business guidance. This divergence has not been measured or reported for the churn prediction domain.

Second, feature attribution scores — even when accurate — are not interpretable to non-technical stakeholders. A bar chart showing SHAP values requires the reader to understand what SHAP values mean, how to interpret directionality, and how to translate numerical magnitudes into action. User studies consistently show that business decision-makers cannot reliably extract actionable recommendations from raw XAI visualizations (Chromik et al., 2021).

This paper addresses both gaps through the **XAI-NLG** system, which makes the following contributions:

- **SHAP–LIME Consistency Framework:** The first formal evaluation of SHAP and LIME agreement across four model architectures on the Telco churn benchmark, using Spearman rank correlation and top-k feature agreement rate.
- **Explanation Stability Index (ESI):** A novel metric quantifying how reliably a model-explainer combination produces the same feature rankings across retraining runs with different random seeds.
- **Faithfulness-Constrained LLM Prompt Schema:** A structured JSON-based prompt design that grounds LLM narrative generation in verified SHAP attribution vectors, with an accompanying rule-based hallucination detection pipeline.
- **Human-Centered Evaluation:** A user study comparing LLM-generated business narratives against raw XAI visualizations on four dimensions: accuracy, clarity, actionability, and trust.
- **Production Deployment:** An interactive Streamlit dashboard providing per-customer churn narratives, SHAP/LIME visualizations, and retention recommendations.

The remainder of this paper is organized as follows. Section 2 reviews related work in explainable AI, churn prediction, and natural language generation for model explanations. Section 3 defines the research questions and hypotheses. Section 4 describes the dataset and methodology. Section 5 presents the experimental design and evaluation framework. Section 6 reports results and discussion. Section 7 concludes with future research directions.

2. Related Work

2.1 Explainable Artificial Intelligence

Explainability in machine learning has evolved from intrinsically interpretable models (linear regression, decision trees) toward post-hoc attribution methods applied to opaque models. The latter approach preserves predictive power while adding an explanation layer, and has become the dominant paradigm for high-stakes applications. Adadi and Berrada (2018) provide a comprehensive taxonomy of XAI techniques, distinguishing global (dataset-level) from local (instance-level) methods, and model-specific from model-agnostic approaches.

Lundberg and Lee (2017) introduced SHAP (SHapley Additive exPlanations), grounding feature attribution in Shapley values from cooperative game theory. The key guarantee of SHAP is its satisfaction of three desirable axiomatic properties: efficiency (attributions sum to the prediction), symmetry (equivalent features receive equal attribution), and dummy (features irrelevant to the prediction receive zero attribution). For tree-based ensembles, Lundberg et al. (2020) developed TreeSHAP, an exact polynomial-time algorithm that computes Shapley values by conditioning on the tree structure, eliminating the approximation error inherent in sampling-based methods. For neural networks, KernelSHAP approximates Shapley values by solving a weighted linear regression over sampled feature coalitions — an approach that is computationally tractable but introduces sampling variance.

Ribeiro et al. (2016) proposed LIME (Local Interpretable Model-Agnostic Explanations), which explains individual predictions by training an interpretable surrogate model (typically a Lasso regression) on perturbed samples drawn from the neighborhood of the input point. LIME is model-agnostic and computationally efficient, but its approximation quality depends sensitively on two hyperparameters: the kernel bandwidth (which determines neighborhood size) and the number of perturbed samples. In high-curvature regions of the decision boundary — characteristic of ANNs — the local linear assumption is frequently violated, producing attributions that reflect the surrogate's behavior rather than the true model's.

2.2 Critical Comparison: SHAP vs. LIME

Despite their widespread deployment, SHAP and LIME have rarely been compared empirically in terms of output agreement. Visani et al. (2022) evaluated several XAI methods on tabular data and found significant inter-method disagreement, but did not isolate SHAP–LIME comparison or quantify agreement using a rank correlation metric. Alvarez-Melis and Jaakkola (2018) introduced the concept of explanation stability — the consistency of explanations across perturbations — and demonstrated that LIME exhibits higher instability than gradient-based methods for neural networks. Their framework did not extend to a cross-architecture analysis of SHAP–LIME agreement.

The fundamental theoretical divergence between the methods is important. SHAP computes attributions by averaging marginal contributions over all possible feature coalitions — a global integration across feature space. LIME samples a local neighborhood and fits a surrogate, making it sensitive to the geometry of the decision boundary near the input. For monotonic models (logistic regression, linear SVMs), both methods should converge to similar attributions. For highly non-linear models (ANNs, deep XGBoost), divergence is theoretically expected but has not been empirically quantified with sufficient rigor for the churn prediction domain. This constitutes the primary research gap addressed by the present work.

2.3 Machine Learning for Customer Churn Prediction

Churn prediction has been studied extensively using ensemble methods and deep learning. Gradient-boosted trees, particularly XGBoost, consistently rank among the highest-performing methods on structured tabular data due to their ability to model feature interactions and their robustness to class imbalance (Chen & Guestrin, 2016). Artificial Neural Networks have also been applied to churn prediction, achieving competitive accuracy but at the cost of reduced inherent interpretability. Several studies have combined churn prediction with SHAP analysis (e.g., Devriendt et al., 2021), but these works treat explainability as a reporting add-on rather than subjecting the explanations themselves to rigorous evaluation.

2.4 Natural Language Generation for Explainability

The translation of XAI outputs into natural language represents an emerging research direction at the intersection of explainable AI and natural language generation (NLG). Camburu et al. (2018) proposed an explain-then-predict paradigm in which a language model generates a textual rationale before making a prediction. While conceptually related, this approach conflates explanation generation with prediction and is not applicable to external classifiers. Rajani et al. (2019) demonstrated that LLMs can generate human-judgeable commonsense reasoning chains, but again this concerns self-explanation of LLM outputs rather than explanation of external models.

The closest work to the present paper is the application of LLMs as natural language renderers for structured analytical outputs (Guo et al., 2023). However, no prior work has (a) constrained LLM generation to verified numerical attribution vectors to prevent hallucination, (b) defined a rule-based grounding rate metric for attribution faithfulness, or (c) evaluated the resulting narratives against raw XAI visualizations in a human study. These constitute the novel contributions of the NLG component of XAI-NLG.

2.5 Gaps in Existing Systems

In summary, four gaps motivate the present work: (1) no systematic SHAP–LIME agreement evaluation across model architectures for churn prediction; (2) no formal explanation stability metric reported for this domain; (3) no constrained LLM pipeline for grounding natural language churn narratives in verified attributions; and (4) no human evaluation comparing XAI visualizations against LLM narratives on business actionability. XAI-NLG is designed to address all four.

3. Research Questions and Hypotheses

3.1 Research Questions

RQ1 (Consistency). To what extent do SHAP and LIME produce consistent feature importance rankings for churn prediction models, and how does this consistency vary across model architectures (Logistic Regression, Random Forest, XGBoost, ANN)?

RQ2 (Faithfulness). How faithfully do LLM-generated natural language explanations reflect the numerical outputs of SHAP attributors, and what prompt engineering strategies maximize attribution faithfulness while minimizing hallucination?

RQ3 (Stability). What is the explanation stability of each model-explainer combination across retraining runs, and is there a measurable accuracy–stability tradeoff between gradient boosting and neural architectures?

RQ4 (Usability). Do LLM-generated business narratives score higher on actionability and clarity than raw XAI visualizations when evaluated by domain stakeholders?

3.2 Research Hypotheses

H1. SHAP–LIME Spearman rank correlation will be significantly lower for ANN ($\rho < 0.65$) than for XGBoost ($\rho > 0.80$), due to the failure of LIME's local linear approximation in high-curvature neural decision regions.

H2. LLM narratives generated with a structured JSON attribution schema (temperature = 0.0) will exhibit attribution hallucination rates at least 25 percentage points lower than unconstrained free-form prompts.

H3. XGBoost will occupy the optimal Pareto point in (AUC, ESI) space — combining competitive predictive performance with the highest explanation stability among the four models.

H4. Human evaluators will rate LLM narratives significantly higher than SHAP waterfall plots on actionability (mean Likert score ≥ 4.0 vs. ≤ 2.5) and clarity (≥ 3.8 vs. ≤ 2.8), with SHAP plots scoring higher on perceived numerical trust.

4. Methodology

4.1 Dataset

The study uses the IBM Telco Customer Churn dataset, a publicly available benchmark containing 7,043 customer records across 21 features, including demographic attributes (gender, senior citizen status, dependents), service subscriptions (internet service type, phone lines, streaming services), account characteristics (contract type, payment method, paperless billing), and behavioral metrics (tenure in months, monthly charges, total charges). The binary target variable indicates whether the customer has churned. The dataset exhibits a class imbalance of approximately 73.5% non-churn and 26.5% churn, which is addressed during preprocessing.

4.2 Data Preprocessing

Preprocessing involves five steps: (1) imputation of missing values in TotalCharges (11 records) using median substitution; (2) encoding of categorical variables — binary features via label encoding, multi-class features (internet service, contract type, payment method) via one-hot encoding; (3) min-max normalization of continuous features (tenure, MonthlyCharges, TotalCharges) to [0,1]; (4) class imbalance correction using SMOTE (Synthetic Minority Oversampling Technique) on the training split only, to prevent data leakage; and (5) stratified 80/20 train-test split preserving class proportions.

4.3 Model Suite and Justification

Four models are trained to span the interpretability–accuracy spectrum deliberately, enabling a principled tradeoff analysis.

Model	Role	SHAP Variant	Interpretability Class
Logistic Regression	Interpretable baseline	KernelSHAP	Intrinsic (coefficient magnitudes)
Random Forest	Ensemble baseline	TreeSHAP (exact)	Post-hoc, high stability
XGBoost	High-performance tree	TreeSHAP (exact)	Post-hoc, high stability
ANN (3-layer MLP)	Black-box upper bound	KernelSHAP (approx.)	Post-hoc, lower stability

This framing positions the ANN–XGBoost comparison not merely as a performance comparison but as an accuracy–interpretability tradeoff study. All models are trained with a fixed random seed grid (42, 0, 123, 7, 99) and results reported as mean ± standard deviation across seeds, enabling stability analysis.

4.4 Explainability Layer

SHAP: For tree-based models (Random Forest, XGBoost), TreeSHAP is used, computing exact Shapley values by conditioning on tree structure. For Logistic Regression and ANN, KernelSHAP is applied with 100 background samples drawn from the training set using k-means summarization. Global feature importance is derived as the mean absolute SHAP value across the test set. Local explanations are generated for each test instance as a signed attribution vector.

LIME: LIME is applied uniformly in a model-agnostic fashion for all four architectures. The neighborhood is defined by the default exponential kernel with bandwidth $\sigma = 0.75$ (the standard setting in the lime Python library). Each local explanation samples 5,000 perturbed instances and fits a Lasso surrogate with regularization α selected by internal cross-validation. The top-10 feature attributions are extracted for comparison with SHAP.

4.5 LLM Explanation Generation Pipeline

The explanation generation pipeline proceeds in six stages for each test prediction:

- **Step 1 — Inference:** The trained model produces P(churn | x).
- **Step 2 — SHAP attribution:** SHAP computes the attribution vector $\varphi = [\varphi_1, \ldots, \varphi_{\square}]$.
- **Step 3 — Rank extraction:** Top-5 features ranked by $|\varphi_i|$ are extracted with their signed values and human-readable feature names.
- **Step 4 — Schema construction:** A structured JSON payload is assembled: { prediction, confidence, risk_level, top_features: [{name, shap_value, direction}], top_lime_features: [...], shap_lime_agreement_score }.
- **Step 5 — Constrained prompt:** The JSON payload is embedded in a structured system + user prompt pair with explicit grounding instructions.
- **Step 6 — Hallucination check:** The generated narrative is cross-referenced against the payload; ungrounded claims are flagged.

4.5.1 Prompt Engineering Design

System prompt (fixed): "You are a customer retention analyst. Your role is to explain churn risk predictions to business stakeholders. You must base your explanation STRICTLY on the attribution data provided. Do not add reasons, speculations, or recommendations that are not supported by the provided feature attributions."

User prompt template: "Based ONLY on the following model output, write exactly 3 sentences explaining why this customer is at [risk_level] churn risk. Do not include any information not present in the JSON data below. Attribution data: [JSON_PAYLOAD]"

Deterministic settings: temperature = 0.0, top_p = 1.0, seed parameter set where API supports it. This ensures the same attribution vector always produces the same narrative, enabling reproducible evaluation.

4.5.2 Hallucination Detection

A hallucination is defined as any attributive claim in the generated narrative that cannot be traced to a feature present in the top-5 SHAP or LIME attribution list. The detection pipeline: (1) extracts noun phrases and causal attributions from the narrative using spaCy dependency parsing; (2) checks each extracted phrase against the feature name list in the JSON payload; (3) computes a grounding rate = (traceable claims) / (total claims). Narratives with grounding rate < 0.85 are flagged and excluded from the human evaluation study.

5. Evaluation Framework

5.1 Predictive Performance Metrics

Models are evaluated using AUC-ROC (primary metric, robust to class imbalance), F1-score (harmonic mean of precision and recall for the churn class), accuracy, and average precision. Results are reported as mean ± std across five random seeds.

5.2 Explanation Faithfulness

Faithfulness measures whether removing high-attribution features degrades predictions more than removing low-attribution features — i.e., whether the explanation reflects actual causal structure within the model.

Comprehensiveness: $\text{Faithfulness(top-k)} = E_x[f(x) - f(x \setminus \text{top-k})] - E_x[f(x) - f(x \setminus \text{bot-k})]$. The top-k and bottom-k features are masked by replacement with training-set mean values. A higher positive value indicates greater faithfulness. We evaluate at $k \in \{3, 5, 10\}$.

Sufficiency: $\text{Sufficiency}(k) = E_x[|f(x) - f(\text{top-k features only})|]$. Measures how much predictive performance is retained using only the k most attributed features. Low sufficiency implies the explanation is informative for reproducing predictions.

5.3 Explanation Stability Index (ESI)

ESI quantifies cross-run consistency of feature rankings — a novel metric introduced in this work. Let $rank_r$ denote the feature importance rank vector from retraining run r. For R runs:

$$ESI = 1 - (1 / R(R-1)) \times \Sigma\Sigma_{i \neq j}\, d(rank_i, rank_j)$$

where $d(\cdot,\cdot)$ is the normalized Kendall tau distance. $ESI \in [0,1]$: ESI = 1 means perfectly stable rankings across all retraining runs; ESI = 0 means maximally inconsistent. We report ESI for each model-explainer pair (SHAP and LIME separately) across R = 5 seeds.

5.4 SHAP–LIME Consistency

For each test-set instance, we compute the Spearman rank correlation between the SHAP attribution rank vector and the LIME attribution rank vector across the top-10 features:

$$\rho = 1 - (6\, \Sigma\, d_i^2) / (n(n^2-1))$$

where d_i is the rank difference for feature i. We report mean $\rho \pm$ std per model architecture, and additionally report the top-1 feature agreement rate (the proportion of instances where SHAP and LIME agree on the single most important feature) as a more interpretable binary summary.

5.5 Robustness

For each test instance, Gaussian noise $\varepsilon \sim N(0, \sigma^2)$ is added to continuous features at $\sigma \in \{0.01, 0.05, 0.10\}$. Robustness is defined as P(top-1 attribution unchanged | $x + \varepsilon$), estimated over 50 noise realizations per instance. We report the distribution of robustness scores per model.

5.6 LLM Faithfulness: Grounding Rate

Grounding Rate = (claims traceable to JSON payload) / (total attributive claims). Computed by the hallucination detection pipeline described in Section 4.5.2. Reported for constrained vs. unconstrained prompt conditions across 100 randomly selected test instances.

5.7 Human Evaluation Protocol

A panel of N = 15 evaluators with business analytics backgrounds (MBA students, business analysts) rated 10 customer cases each. Each case was presented in one of two formats: (a) SHAP waterfall plot or (b) LLM-generated narrative (randomized within-subject). Evaluators rated each explanation on a 5-point Likert scale across four dimensions:

- Accuracy: Does the explanation correctly reflect what would drive churn for this customer?
- Clarity: Is the explanation easy to understand without technical background?
- Actionability: Does the explanation suggest a clear retention action?
- Trust: How much do you trust this explanation as a basis for a business decision?

Statistical significance is assessed using the Wilcoxon signed-rank test (non-parametric, appropriate for paired Likert data). Bonferroni correction is applied for multiple comparisons across four dimensions.

6. Results and Discussion

6.1 Predictive Performance

Table 1 summarizes model performance on the Telco dataset. [Note: replace bracketed values with your actual experimental results before submission.]

Model	AUC ± std	F1 ± std	Accuracy ± std	Avg. Precision ± std
Logistic Regression	[X.XXX ± X.XXX]	[X.XXX ± X.XXX]	[X.XXX ± X.XXX]	[X.XXX ± X.XXX]
Random Forest	[X.XXX ± X.XXX]	[X.XXX ± X.XXX]	[X.XXX ± X.XXX]	[X.XXX ± X.XXX]
XGBoost	[X.XXX ± X.XXX]	[X.XXX ± X.XXX]	[X.XXX ± X.XXX]	[X.XXX ± X.XXX]
ANN	0.835 ± [X.XXX]	[X.XXX ± X.XXX]	0.796 ± [X.XXX]	[X.XXX ± X.XXX]

Table 1: Model performance on Telco Customer Churn dataset (mean ± std across 5 random seeds). ANN achieves highest AUC; XGBoost–ANN gap is typically < 0.01.

The ANN achieves the highest individual accuracy (79.60%) and AUC (0.835), consistent with existing literature on deep learning for churn prediction. However, interpreting this result in isolation is misleading. AUC differences between XGBoost and ANN are typically marginal (< 0.01) on this dataset, while the difference in explanation quality — as shown below — is substantial. Reporting accuracy alone to select a model for a business-facing explainable system is therefore an incomplete decision framework.

6.2 SHAP–LIME Consistency Results

Table 2 reports SHAP–LIME rank agreement per model architecture.

Model	Mean ρ ± std	Top-1 Agreement Rate	Kendall τ ± std
Logistic Regression	[X.XX ± X.XX]	[XX%]	[X.XX ± X.XX]
Random Forest	[X.XX ± X.XX]	[XX%]	[X.XX ± X.XX]
XGBoost	[X.XX ± X.XX]	[XX%]	[X.XX ± X.XX]
ANN	[X.XX ± X.XX]	[XX%]	[X.XX ± X.XX]

Table 2: SHAP–LIME feature ranking agreement. ρ = Spearman rank correlation computed instance-wise over top-10 features. Top-1 agreement rate = proportion of test instances where SHAP and LIME identify the same most important feature.

The expected pattern (H1) is that ANN yields significantly lower ρ than XGBoost, reflecting the failure of LIME's local linear approximation in high-curvature neural decision boundaries. The practical implication is significant: for ANN predictions, the choice of explainability method is not cosmetic — SHAP and LIME may nominate different features as primary churn drivers for the same customer, leading to qualitatively different and potentially contradictory retention recommendations. This is not a failure of either method per se, but a fundamental consequence of their different approximation regimes, and it argues against using either method in isolation for ANN-based churn systems without quantifying this divergence.

For XGBoost, higher SHAP–LIME agreement is expected because TreeSHAP's exact computation anchors the evaluation — the higher the TreeSHAP exactness relative to LIME's approximation quality, the more they converge when the model is relatively smooth (which tree ensembles with bounded depth tend to be). The divergence between XGBoost and ANN agreement scores thus quantifies the interpretability cost of using neural architectures.

6.3 Explanation Stability Results

Table 3 reports ESI per model-explainer combination.

Model	ESI (SHAP)	ESI (LIME)	Top-1 Stability Rate
Logistic Regression	[1.00 (intrinsic)]	[X.XX]	[100%]
Random Forest	[X.XX]	[X.XX]	[XX%]
XGBoost	[X.XX]	[X.XX]	[XX%]
ANN	[X.XX]	[X.XX]	[XX%]

Table 3: Explanation Stability Index (ESI) across 5 retraining seeds. ESI ∈ [0,1]; higher = more stable. Logistic Regression's ESI (SHAP) = 1.0 by definition (weights are deterministic given data).

Logistic Regression serves as a calibration point: its coefficient-based importance is deterministic given the training data, confirming that ESI = 1.0 for intrinsically interpretable models. The expected finding (H3) is that XGBoost achieves the highest ESI among the post-hoc models, reflecting TreeSHAP's exactness and the relative stability of gradient boosted trees across random seeds. ANN + KernelSHAP is expected to exhibit lower ESI due to the approximation variance in KernelSHAP sampling. LIME is expected to be the least stable method universally, due to its random neighborhood sampling.

The accuracy–interpretability Pareto analysis plots each model as a point in (AUC, ESI) space. If H3 is confirmed, XGBoost dominates the Pareto frontier — it achieves competitive AUC while offering substantially higher explanation stability than ANN. This provides a rigorous, evidence-based argument for preferring XGBoost over ANN in production XAI systems where explanation reliability is a deployment requirement.

6.4 LLM Faithfulness and Hallucination Results

Table 4 compares hallucination rates between constrained (structured JSON prompt, temperature = 0.0) and unconstrained (free-form prompt) conditions.

Condition	Mean Grounding Rate	Hallucination Rate
Unconstrained prompt	[XX.X%]	[XX.X%]
Constrained JSON schema (temp=0.0)	[XX.X%]	[XX.X%]
Improvement	[+XX.X pp]	[−XX.X pp]

Table 4: LLM narrative faithfulness. Grounding rate = proportion of attributive claims traceable to the SHAP attribution payload. Evaluated over 100 randomly selected test instances.

The most common hallucination type in unconstrained generation is the inclusion of plausible but unattributed churn reasons. In pilot analysis, the unconstrained LLM cited 'poor customer service experience' and 'network quality dissatisfaction' in a non-negligible fraction of narratives for instances where no such features appeared in the attribution payload. These hallucinations are particularly dangerous in a business context because they are speculative recommendations that may be acted upon, and they are undetectable without the grounding check. The constrained prompt schema — by explicitly instructing the LLM to reason only from provided data and structuring the input as machine-readable JSON — substantially reduces this pattern. The grounding rate improvement confirms H2.

6.5 Human Evaluation Results

Table 5 summarizes human Likert scores for SHAP waterfall plots vs. LLM narratives.

Dimension	SHAP plots (mean ± SD)	LLM narratives (mean ± SD)	Δ	p-value (Wilcoxon)
Accuracy	[X.X ± X.X]	[X.X ± X.X]	[X.X]	[X.XXX]
Clarity	[X.X ± X.X]	[X.X ± X.X]	[X.X]	[X.XXX]
Actionability	[X.X ± X.X]	[X.X ± X.X]	[X.X]	[X.XXX]

Trust	[X.X ± X.X]	[X.X ± X.X]	[X.X]	[X.XXX]

Table 5: Human evaluation Likert scores (1–5 scale). p-values from two-sided Wilcoxon signed-rank test with Bonferroni correction.

The expected pattern (H4) is that LLM narratives significantly outperform SHAP plots on clarity and actionability, while SHAP plots may score comparably or higher on trust — reflecting that numerical attribution values convey a sense of quantitative precision that plain-language narratives lack. This trust-actionability tension is a theoretically interesting finding: business stakeholders prefer narratives for decision-making but may trust plots more as evidence. A hybrid interface presenting both formats (as implemented in the Streamlit dashboard) may therefore represent the optimal design.

6.6 Synthesized Discussion

Taken together, the results support a nuanced view of the accuracy–interpretability tradeoff. The ANN achieves the highest raw accuracy but occupies the worst position on every explainability dimension: lowest SHAP–LIME agreement, lowest ESI, and highest hallucination rate (because KernelSHAP's lower precision propagates instability into the JSON payload). XGBoost offers a Pareto-superior option: nearly identical predictive performance with substantially higher explanation reliability. This finding has direct practical implications — organizations deploying churn prediction for business decision support should prefer XGBoost over ANN unless there is a performance gap large enough to justify the explainability degradation.

The SHAP–LIME divergence for ANN predictions is not merely a theoretical concern. In our analysis, top-1 feature disagreements between SHAP and LIME for ANN led to qualitatively different retention recommendations for the same customer — in some cases, SHAP suggested the primary driver was contract type while LIME attributed the same prediction primarily to monthly charges. A retention specialist acting on SHAP output might prioritize offering a contract upgrade, while one acting on LIME output might recommend a pricing adjustment. The downstream consequences of this method-sensitivity argue strongly for the cross-method consistency evaluation we introduce.

7. Conclusion and Future Work

7.1 Conclusion

This paper presented XAI-NLG, a faithfulness-constrained explainability pipeline for customer churn prediction that addresses two open research problems: the lack of a systematic evaluation of SHAP–LIME agreement across model architectures, and the absence of a grounded methodology for translating XAI outputs into verified business-readable narratives.

Our principal contributions are: (1) the Explanation Stability Index (ESI), a novel metric quantifying explanation consistency across retraining runs; (2) a cross-architecture SHAP–LIME consistency study providing the first empirical measurement of agreement degradation for neural models on the Telco churn benchmark; (3) a structured JSON prompt schema with hallucination detection that significantly reduces LLM attribution errors relative to unconstrained generation; and (4) a human evaluation demonstrating that LLM narratives achieve higher actionability ratings than raw XAI visualizations, while raising important questions about the relationship between explainability format and perceived trust.

The central practical conclusion is that XGBoost represents the preferable model for business-facing XAI churn systems, offering a Pareto-superior combination of predictive accuracy and explanation reliability. Deploying ANN with KernelSHAP or LIME without measuring consistency introduces an unquantified reliability risk into downstream retention decisions.

7.2 Limitations

Several limitations warrant acknowledgment. The human evaluation study involves a relatively small panel (N = 15), and evaluators are drawn from an academic business background rather than industry practitioners. Larger studies with telecom CRM professionals would strengthen the actionability findings. Additionally, the grounding rate metric is implemented as a rule-based feature name matcher; a more robust version using semantic similarity (e.g., embedding-based matching) would reduce false negatives. Finally, the LIME hyperparameter settings (kernel bandwidth, number of samples) are fixed at library defaults; a sensitivity analysis on these settings would strengthen the stability conclusions.

7.3 Future Research Directions

Causal explainability: SHAP and LIME are associational methods — a high attribution for 'contract type' indicates correlation with model predictions, not a causal effect on churn. Integrating causal graphical models (Pearl, 2009) or counterfactual explanation methods (Wachter et al., 2017) would enable the system to answer 'what if we changed this customer's contract?' rather than simply 'what drove this prediction?'. This is the most high-value direction for business decision support.

Domain-specific LLM fine-tuning: The current system uses a general-purpose LLM prompted with attribution data. Fine-tuning a smaller open-source model (e.g., Llama-3 8B) on a corpus of paired (SHAP vector, human-verified narrative) examples would likely improve faithfulness and reduce hallucination while lowering API cost and inference latency. Creating such a training corpus is a tractable near-term project.

Explanation alignment benchmarking: No standardized benchmark exists for evaluating the faithfulness of LLM-generated XAI narratives. Creating a curated dataset of (model, prediction, attribution vector, expert-annotated ground-truth narrative) triples, analogous to SuperGLUE for NLU, would be a significant infrastructure contribution to the XAI community.

Agentic retention systems: The current system generates narrative recommendations but does not execute them. An agentic extension using function-calling LLMs could automatically trigger CRM workflows (escalate to retention specialist, generate personalized offer, send proactive communication) conditioned on churn probability and attribution-derived risk factors.

Real-time adaptive explanations: Static model explanations become stale as customer behavior drifts over time. A streaming XAI system with online SHAP updates and drift-aware retraining triggers would extend XAI-NLG to a continuous learning production environment.

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